Task 2:
Q:- Write a single cin statment
that read's input into

```
int age;
double pay;
char section;
```

```
#include <iostream>
using namespace std;
int main() {

 int age;
double pay;
char section;
 cin >> age >> pay >> section;
// sigle cin statment.
cout << age << "20" << pay << "200" <<
<< section << endl;
```

Lab 3 - Task 2: Single cin statement

```
#include <iostream>
using namespace std;
int main() {
    int age;
    double pay;
    char section;
    cin >> age >> pay >> section;   // single cin statement
    cout << age << " " << pay << " " << section << endl;
    return 0;
}
```

Run 1



Task 2

Q: Write the value of each expression according to C++ rules.

```
#include <iostream>
using namespace std;

int main() {
cout << 28 / 4 - 2 << endl;          // 5
cout << 6 + 12 * 2 - 8 << endl;      // 22
cout << 4 + 8 * 2 << endl;           // 20
cout << 6 + 17 % 3 - 2 << endl;      // 6
cout << 2 + 22 * (9 - 7) << endl;    // 46
cout << (8 + 7) * 2 << endl;         // 30
cout << (16 + 7) % 2 - 1 << endl;    // 0
cout << 12 / (10 - 6) << endl;       // 3
cout << (19 - 3) * (2 + 2) / 4 <<
endl;                                // 16
```

Table values

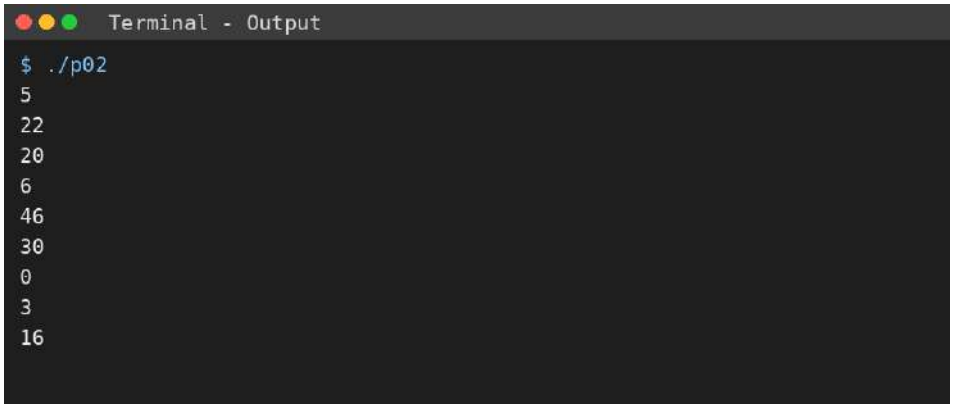
5, 22, 20, 6, 46, 30, 0, 3, 16

Lab 3 - Task 2: Expression values

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << 28 / 4 - 2 << endl;
    cout << 6 + 12 * 2 - 8 << endl;
    cout << 4 + 8 * 2 << endl;
    cout << 6 + 17 % 3 - 2 << endl;
    cout << 2 + 22 * (9 - 7) << endl;
    cout << (8 + 7) * 2 << endl;
    cout << (16 + 7) % 2 - 1 << endl;
    cout << 12 / (10 - 6) << endl;
    cout << (19 - 3) * (2 + 2) / 4 << endl;
    return 0;
}
```

Run 1





Lab 3
Task 3, 4, 5

Task 3
Answer is C units is promoted to float, mass stays float, mass * unit; is float and then it is converted to double for weight

Task 4
Answer is B (8)
B * C = 2 * 4.2 = 8.4, But a is int so the decimal part is cut and a = 8.

Task 5.
unit's each = static_cast<double>(qty)
/ salesReps;

Task 6

Math Tutor

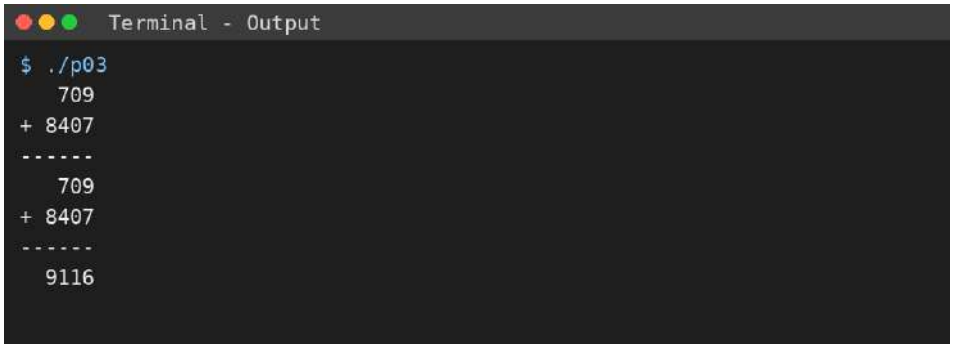
```
#include <iostream>
#include <iomanip>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
srand(time(0));
int num1 = rand() % 9000 + 1000;
int num2 = rand() % 9000 + 1000;
cout << setw(6) << num1 << endl;
cout << "+" << setw(5) << num2
<< endl;
cout << "------" << endl;
cin.get();
cout << setw(6) << num1 << endl;
cout << "+" << setw(5) << num2
<< endl;
cout << "------" << endl;
cout << setw(6) << num1 + num2
<< endl;
return 0;
}
```

Lab 3 - Task 6: Math Tutor

```
#include <iostream>
#include <iomanip>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
   srand(time(0));
   int num1 = rand() % 900 + 100;
   int num2 = rand() % 9000 + 1000;
   cout << setw(6) << num1 << endl;
   cout << "+" << setw(5) << num2 << endl;
   cout << "------" << endl;
   cin.get();
   cout << setw(6) << num1 << endl;
   cout << "+" << setw(5) << num2 << endl;
   cout << "------" << endl;
   cout << setw(6) << num1 + num2 << endl;
   return 0;
}
```

Run 1





Home Task

Program 1

```
using namespace std;
int main() {
 double number1, number2, sum;
cout << enter a number: ";
 cin >> number1;
 cout << " Enter another number: ";
 cin >> number2;
 sum = number1 + number2;
 cout << " The sum is " << sum <<
 endl;
 return 0;
}
```

Output The sum is 20

Program 2

```
# include <iostream>
using namespace std;
 int main() {
int number1, number2;
   float quotient
cout << "Enter two numbers: ";
```

```
cin >> number1 >> number2;
quotient = static_cast<float>(
number2)
    / number2;
cout << quotient;
}
```

Home Task 2

Average

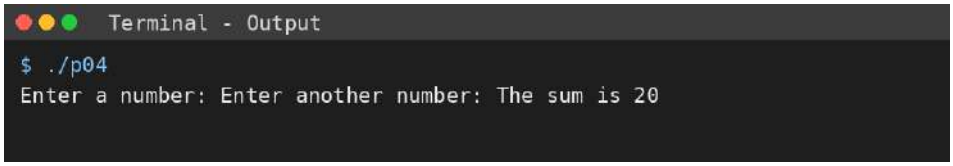
```
#include <iostream>
using namespace std;
int main () {
int a=28, b=32, c=37
d=24, e=33;
int sum = a+b+c+d+e;
double avg = sum/5.0;
cout << "Average =" <<avg; //30.8
```

Home Task - Program 1: Sum of two numbers

```
#include <iostream>
using namespace std;
int main() {
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
    cout << "The sum is " << sum << endl;
    return 0;
}
```

Run 1





Home Task - Program 2: Quotient (static_cast)

```
#include <iostream>
using namespace std;
int main() {
    int number1, number2;
    float quotient;
    cout << "Enter two numbers: ";
    cin >> number1 >> number2;
    quotient = static_cast<float>(number1) / number2;
    cout << quotient;
    return 0;
}
```

Run 1





Run 2





Home Task 2: Average

```cpp
#include <iostream>
using namespace std;
int main() {
    int a = 28, b = 32, c = 37,
        d = 24, e = 33;
    int sum = a + b + c + d + e;
    double avg = sum / 5.0;
    cout << "Average = " << avg;   // 30.8
    return 0;
}
```

Run 1



Lab4-

Task 2

```
#include <iostream>
using namespace std;
int main () {
 int n;
 cin >> n;
 if >>n;
 ,if (n >0)
 cout << "positive: " <<n;
else
 cout << "Negative";
}
```

Task 2

```
#include <iostream>
using namespace std;
int main(){
int n;
 cin >>n;
if (n>0) cout << "Positive";
else if (n<0) cout << "Negative";
else cout << "Zero";
```

Lab 4 - Task 1: Positive / Negative

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    if (n > 0)
        cout << "Positive: " << n;
    else
        cout << "Negative";
    return 0;
}
```

Run 1





Run 2





Lab 4 - Task 2: Positive / Negative / Zero

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    if (n > 0) cout << "Positive";
    else if (n < 0) cout << "Negative";
    else cout << "Zero";
    return 0;
}
```

Run 1



Run 2





Run 3





Task 3

Grade

```
#include <iostream>
using namespace std;
int main() {
double mark;
cin >> mark;
if (mark >= 75) cout << "A";
else if (mark >= 60) cout << "B";
else if (mark >= 45) cout << "C";
else cout << "D";
}
```

Lab 4 - Task 3: Grade

```cpp
#include <iostream>
using namespace std;
int main() {
    double mark;
    cin >> mark;
    if (mark >= 75) cout << "A";
    else if (mark >= 60) cout << "B";
    else if (mark >= 45) cout << "C";
    else cout << "D";
    return 0;
}
```

Run 1





Run 2





Run 3





Run 4




Terminal - Output
$ ./p09
D

Task 4a
(largest)

```
#include <iostream>
using namespace std;
int main () {
 int a, b, c;
 cin >> a >> b >> c;
 if (a >= b) {
 if (a >= b) {
 if (a >= c) cout << a;
  else cout << c;
 }
 else {
  if (b >= c) cout << b;
 else cout << c;
  }
 }
```

Task 4B
(Digits)

```
" " "
" " "
int main () {
int n , d = 0;
 cin >> n;
```